Guanyu Hu

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RESEARCH INTERESTS

Trustworthy AI (Fairness, Accountability, Transparency & Ethics)
Fairness-Oriented Dataset & Evaluation Protocol Design; Bias Mitigation in MTL and VLMs; Fair Domain Generalization.

Affective Computing (Interpretable, Multimodal & Multi-task)
Causal Learning; Cognitively Aligned Representation Learning; Multi-Task Learning (MTL); Domain Adaptation.

WORK EXPERIENCE & EDUCATION

HIIT Postdoctoral Fellow | *Aalto University* — Helsinki, Finland
Three-Year Research Fellowship (Fully Funded) | *Supervisor: Prof. Guoying Zhao & Juho Kannala* — Oct. 2026 - Present

Research Associate | *Queen Mary University of London* — London, UK
AI-Enabled Face Analysis in-the-Wild | *Supervisor: Prof. Dimitrios Kollias* — Oct. 2024 - Oct. 2026

Visiting Ph.D. | *Queen Mary University of London* — London, UK
Innovative Leading Talents Scholarship | *Supervisor: Prof. Dimitrios Kollias* — Oct. 2023 - Oct. 2024

Ph.D. | *Xi'an Jiaotong University* — Xi'an, China
Artificial Intelligence | *Supervisor: Prof. Xinyu Yang* — Sep. 2020 - Mar. 2026

M.Sc. | *Xi'an Jiaotong-Liverpool University* — Suzhou, China
Applied Informatics — Sep. 2018 - Apr. 2020

B.Sc. | *Zhengzhou University* — Zhengzhou, China
Computer Science — Sep. 2013 - Jun. 2017

PUBLICATIONS

[1] ***NeurIPS 2026*** | ***Guanyu Hu***, *Tangzheng Lian, Na Yan, Dimitrios Kollias, Xinyu Yang, Oya Celiktutan, Siyang Song, Zeyu Fu.*
FairMT: Fairness for Heterogeneous Multi-Task Learning.

[2] ***NeurIPS 2026*** | ***Guanyu Hu***, *Tangzheng Lian, Dimitrios Kollias, Oya Celiktutan, Xinyu Yang.*
CausalAffect: Causally Guided Learning of Psychology-Aligned Facial Affect Relations.

[3] ***ICML 2026*** | ***Guanyu Hu***, *Dimitrios Kollias, Xinyu Yang.*
AURA: Visually Interpretable Affective Understanding via Robust Archetypes.

[4] ***IEEE/CVF WACV 2026*** | ***Guanyu Hu***, *Dimitrios Kollias, Xinyu Yang.*
From Cognitive Priors to Instance Semantics: A Unified Framework for Multi-task Affective Computing.

[5] ***IEEE/CVF CVPR 2026*** | *Tangzheng Lian,* ***Guanyu Hu***, *Yijing Ren, Dimitrios Kollias, Oya Celiktutan.*
A Closed-Form Solution for Debiasing Vision-Language Models with Utility Guarantees Across Modalities and Tasks.

[6] ***AAAI 2026 (Oral)*** | *Tangzheng Lian,* ***Guanyu Hu***, *Dimitrios Kollias, Xinyu Yang, and Oya Celiktutan.*
Fair Domain Generalization: An Information-Theoretic View.

[7] ***ECCV 2026*** | *D. Kollias, S. Zafeiriou, I. Kotsia, E. Granger, M. Pedersoli, A. L. Koerich, S. L. Bacon, O. Celiktutan, ...,* ***Guanyu Hu***.
From Complex Behavior to Intelligent Human-Centered AI: The 11th ABAW Workshop & Competition.

[8] ***IEEE/CVF CVPR 2026*** | *D. Kollias, P. Tzirakis, A. Cowen, S. Zafeiriou, I. Kotsia, E. Granger, M. Pedersoli, S. Bacon, J. Madsen, ...,* ***Guanyu Hu***.
From Affect to Complex Behavior: Advancing Multimodal Human-Centered AI at 10th ABAW Workshop & Competition.

[9] ***ACM MM 2025 (Oral)*** | ***Guanyu Hu***, *Dimitrios Kollias, Xinyu Yang.*
Grounding Emotion Recognition with Visual Prototypes: VEGA - Revisiting CLIP in MERC.

[10] ***IEEE Transactions on Biometrics, Behavior, and Identity Science (T-BIOM)*** | ***G. Hu***, *D. Kollias, E. Papadopoulou, P. Tzouveli, J. Wei, X. Yang.*
Rethinking Affect Analysis: A Protocol for Ensuring Fairness and Consistency.

[11] ***IEEE/CVF ICCV 2025*** | *D. Kollias, S. Zafeiriou, I. Kotsia, G. Slabaugh, D. Senadeera, J. Zheng, K. Kumar K. Yadav, C. Shao,* ***Guanyu Hu***.
From Emotions to Violence: Multimodal Fine-Grained Behavior Analysis at 9th ABAW.

[12] ***IEEE/CVF CVPR 2025*** | *D. Kollias, P. Tzirakis, A. Cowen, S. Zafeiriou, I. Kotsia, E. Granger,* ***Guanyu Hu*** *et al.*
Advancements in Affective and Behavior Analysis: The 8th ABAW Workshop and Competition.

[13] ***IEEE FG 2024 (Best Reviewed Paper)*** | ***Guanyu Hu***, *Eleni Papadopoulou, Dimitrios Kollias, Paraskevi Tzouveli, Jie Wei, Xinyu Yang.*
Bridging the Gap: Protocol Towards Fair and Consistent Affect Analysis.

[14] ***IEEE IJCB 2024*** | ***Guanyu Hu***, *Jie Wei, Siyang Song, Dimitrios Kollias, Xinyu Yang, Zhonglin Sun, Odysseus Kaloidas.*
Robust Facial Reactions Generation: An Emotion-Aware Framework with Modality Compensation.

[15] ***Expert Systems with Applications (ESA) 2024*** | *Jie Wei,* ***Guanyu Hu***, *Xinyu Yang, Anh Tuan Luu, Yizhuo Dong.*
Learning Facial Expression and Body Gesture Visual Information for Video Emotion Recognition.

[16] ***Journal of the American Medical Informatics Association (JAMIA) 2024*** | *X. Lu, C. Yang, L. Liang,* ***Guanyu Hu***, *Z. Zhong, Z. Jiang.*
Artificial Intelligence for Optimizing Recruitment and Retention in Clinical Trials: A Scoping Review.

[17] ***IEEE/CVF CVPR 2024*** | *D. Kollias, A. Psaroudakis, A. Arsenos, P. Theofilou, C. Shao,* ***Guanyu Hu****, I. Patras.*
MMA-MRNNet: Harnessing Multiple Models of Affect and Dynamic Masked RNN for Precise Facial Expression Intensity Estimation.

[18] ***ECCV 2024*** | *D. Kollias, S. Zafeiriou, I. Kotsia, A. Dhall, S. Ghosh, C. Shao,* ***Guanyu Hu****.*
7th ABAW Competition: Multi-task Learning and Compound Expression Recognition.

[19] ***IEEE/CVF CVPR 2024*** | *D. Kollias, P. Tzirakis, A, Cowen, S, Zafeiriou, I. Kotsia, A. Baird, C. Gagne, C. Shao,* ***Guanyu Hu****.*
The 6th Affective Behavior Analysis In-the-wild (ABAW) Competition.

[20] ***IEEE ICASSP 2023 (Oral)*** | *Jie Wei,* ***Guanyu Hu****, Luu Anh Tuan, Xinyu Yang, and Wenjing Zhu.*
Multi-scale Receptive Field Graph Model for Emotion Recognition in Conversations.

[21] ***INTERSPEECH 2022*** | *Jie Wei,* ***Guanyu Hu****, Xinyu Yang, Anh Tuan Luu, and Yizhuo Dong.*
Audio-Visual Domain Adaptation Feature Fusion for Speech Emotion Recognition.

RESEARCH GRANTS

(1) Finnish Research Impact Foundation (FRIF) — TIA Postdoc Grant | €229,933 2026

Psychology-Guided AI for Better Focus, Motivation, and Engagement in Digital Education | 9.8% success rate

Led the conception and development of the successful grant proposal, shaping the scientific agenda, research methodology, four work packages, and industry-impact strategy. Established the foundation for a 24-month academia–industry research collaboration. (PI: Prof. Guoying Zhao)

(2) HIIT Postdoctoral Fellowship | Three-Year Fully Funded Fellowship 2026

Helsinki Institute for Information Technology (HIIT), Aalto University | 2.44% offer rate

Selected through a highly competitive, research-proposal-based fellowship process, demonstrating the ability to formulate an independent research agenda, develop original research directions, and articulate objectives and expected contributions.

PROJECT EXPERIENCE

As a student group leader, coordinated a 3-year collaboration with Shaanxi Radio & Television (China), delivering **3 projects** and winning **4 awards** in NRTA-hosted national innovation competitions, with key responsibilities including:

- **Led proposal, roadmap, milestones, and resource allocation.**
- **Managed delivery risks and compliance** (e.g., data privacy).
- **Led an 8-member team** (3 PhD and 5 Master' s students) for algorithm development and delivery.

(1) Highlight Short Video Intelligent Auto-editing Platform | £87k | *Student Group Leader* 2020 - 2021

Delivered a multi-camera capture and cloud analytics platform that auto-generates highlight short videos and pushes them to user devices, supporting sports, tourism, and large events. **Designed and optimized multimodal algorithms** (video understanding and highlight ranking), enhancing representations, fusion strategies; enabled a **real-time, parallel analytics pipeline** for scalable multi-camera deployments. **Served 100K+ users** across deployments including the Xiamen Marathon, Xi' an Marathon, Xi' an Museum, and the Mausoleum of the First Qin Emperor. **Awarded Second Prize** in the NRTA Artificial Intelligence Application Innovation Competition in 2021.

(2) Multimodal Analysis and 4K TV Programme Production Platform | £95k | *Student Group Leader* 2021 - 2022

Delivered an end-to-end AI platform for 4K TV content analysis and production. Enabled key capabilities including AI restoration and Ultra-HD enhancement, intelligent editing, automated review, and asset management. **Unified full broadcast production lifecycle coverage,** connecting planning–newsgathering–editing–review–publishing and asset management, enabling content understanding, intelligent editing, quality control, and multi-channel distribution. **Designed and integrated 20+ algorithms**, grouped into scene understanding, restoration/super-resolution, detection, OCR, and identity analytics (re-ID and important-person recognition). **Led the data–model–evaluation design**, from requirement-driven dataset setup to taxonomy configuration, annotation alignment, and quality control. **Awarded Second Prize and Excellence Award** in the NRTA New Video Technology Application Innovation Competition in 2021 and 2022.

(3) Dialect-Enabled Digital TV Presenter Generation Platform | £76k | *Student Group Leader* 2022 - 2023

Delivered a text-driven digital presenter generation platform for automated breaking-news and weather video production, deployed across multiple apps and websites. **Designed and integrated three generation pipelines** (3D reconstruction–based, GAN-based, and NeRF-based), enabling accurate lip–speech synchronization, natural facial expressions, and replaceable broadcast backgrounds. **Streamlined real-time TV content production workflows** for text-to-video news and weather presentation, enabling rapid content generation, improving efficiency, and reducing production costs, while supporting the preservation of classic and valuable on-screen presenter identities. **Awarded Second Prize** in the NRTA New Video Technology Application Innovation Competition (2022).

PATENTS

[1] **An Expression Recognition Method Based on Local Temporal-Spatial and Global Temporal Sequence Features,** 2021, *CN202110925608.5.*

[2] **A Video Emotion Recognition Method Based on Body Gesture Change Representation,** 2021, *CN202110926712.6.*

[3] **A Depression Recognition Method Based on Voice and Facial Emotion Cues,** 2021, *CN202110874297.4.*

[4] **A Speaker Face Synthesis Method, Device and Equipment Based on Generative Adversarial Networks,** 2022, *CN202211493192.5.*

[5] **An Occluded Face Recognition Method, Device and Equipment Based on Multi-Scale Attention Feature Learning,** 2022, *CN202211493911.3*

OPEN-SOURCE DATASETS AND BENCHMARKS

[Towards Fair and Consistent Affect Analysis]

- **Developed a fairness-aware unified protocol** for six affective datasets, addressing inconsistent splits and evaluations.
- **Added demographic annotations, standardized metrics**, and a shared framework for expression, AU, and VA tasks.
- **Released pre-trained models and new leaderboards** to enable fairer, more reproducible benchmarking.

AWARDS

- **Top Reviewer** | *NeurIPS 2026* 2026
- **Best Reviewed Paper** | *IEEE FG 2024* 2024
  Bridging the Gap: Protocol Towards Fair and Consistent Affect Analysis.
- **Innovative Leading Talents Scholarship** | *Xi'an Jiaotong University* 2023
- **Academic Scholarship** | *Xi'an Jiaotong University* 2023
- **Outstanding Student Leader** | *Xi'an Jiaotong University* 2023
- **Outstanding Student** | *Xi'an Jiaotong University* 2022
- **Second Prize** | *China National Radio and Television Administration* 2022
  Artificial Intelligence Application Innovation Competition
  Local Language IP Broadcasting Integration Platform Based on NeRF.
- **Excellence Award** | *China National Radio and Television Administration* 2022
  New Video Technology Application Innovation Competition
  An Interactive E-commerce Media System Based on Video Scene Understanding.
- **National Encouragement Scholarship** | *China Ministry of Education (MoE)* 2016

TEACHING EXPERIENCE

Student Mentoring

- Yixin Song | MSc student | Emotion-aware Talking Head Generation | Xi' an Jiaotong University 2020-2023
- Shuo Zhang | MSc student | Composition-Aware Image Cropping | Xi' an Jiaotong University 2020-2023
- Zhixin Cao | MSc student | Robust Face Recognition with Occlusions | Xi' an Jiaotong University 2020-2023
- Yuliang Hou | MSc student | Shaanxi Dialect–Enabled Speech Generation | Xi' an Jiaotong University 2020-2023
- Peixu Zhang | MSc student | NeRF-based Neural Talking Head Synthesis | Xi' an Jiaotong University 2020-2023

Teaching Assistant

- Graph Theory and Applications | 052130 | Xi' an Jiaotong University 2020, 2021 and 2022
- Data Mining | ECS607U | Queen Mary University of London 2023, 2024 and 2025

Assistant Examiner

- Advanced Network Programming | EBU5042 | Queen Mary University of London 2024 and 2025
- Middleware | EBU6501 | Queen Mary University of London 2024 and 2025
- Software Engineering | EBU6304 | Queen Mary University of London 2024
- Communications Systems | ECS528U | Queen Mary University of London 2024
- Programming Fundamentals for Engineers | ECS429U | Queen Mary University of London 2024

RESEARCH ACTIVITIES

Program Committee(programme/data/session chair) | ABAW Workshop & Competition Series:

- **ECCV 2026** (11th ABAW) From Complex Behavior to Intelligent Human-Centered AI
- **CVPR 2026** (10th ABAW) From Affect to Complex Behavior: Advancing Multimodal Human-centered AI
- **ICCV 2025** (9th ABAW) From Emotions to Violence: Multimodal Fine-Grained Behavior Analysis
- **CVPR 2025** (8th ABAW) Advancements in Affective and Behavior Analysis
- **ECCV 2024** (7th ABAW) Multi-task Learning and Compound Expression Recognition
- **CVPR 2024** (6th ABAW) Affective Behavior Analysis in-the-wild

Reviewer & Program Committee

- **Journal:**

  IEEE Transactions on Pattern Analysis and Machine Intelligence, IEEE Transactions on Affective Computing, Pattern Recognition, Expert Systems with Applications, Neural Networks, International Journal of Computer Vision, Image and Vision Computing, Biomedical Signal Processing and Control, Applied Soft Computing, ACM Transactions on Multimedia Computing, Communications, and Applications

- **Conference:**

  ICLR, ICML, NeurIPS, AAAI, IEEE/CVF CVPR, ICCV, ECCV, IEEE/CVF WACV, ACM MM, BMVC, ACM ICMR, ACM MM Asia